Efficiency and resource measurement

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Declare the workload. Measure completed work. Apply every requirement.

Calculate timing ratios and parameter payloads while keeping resource claims within measured scope.

Keep one habit: Raw timings → derived comparison → quality gates → separate mean-based picture → parameter account → executable audit → independent timings.

MAIN LESSON · SECTION 1

Choose a model that meets quality and resource requirements

> Name the measurement boundary before comparing numbers.

Learning goals

I can calculate **throughput** and amortised time per example from repeated timings.

I can count dense-network parameters and distinguish storage from **peak memory**.

I can compare resource use under fixed quality and measurement conditions.

Retrieve fair comparisons

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Dataset shift in neural models. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Transfer learning. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sequence representations. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Why should two models be evaluated on the same examples and subgroup mix before comparing their accuracy and speed?

> **My starting explanation**

End goal

By the end, I can explain efficiency and resource measurement using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

This week you will practise:

Calculate **throughput** and amortised time per example from repeated timings

Count dense-network parameters and distinguish storage from **peak memory**

Compare resource use under fixed quality and measurement conditions

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Read the timing protocol before comparing the models

> Both candidates process the same cases under the same declared conditions.

Quality and resources together

A neural model can be faster yet fail the task's accuracy requirement. **Training** cost and **prediction** cost are different: **training** updates parameters, while inference uses the fitted model.

Compare quality and resources together under a declared test, rather than calling a model efficient from one small number.

Complete supplied measurement protocol

Use these invented classroom timings for models A and B. Each timed pass **process**es the same 100 pre**process**ed examples sequentially, one at a time, on the same device and precision.

Start before the first **prediction** and stop after the final **prediction** completes. Loading and pre**process**ing are excluded for both. One untimed warm-up pass precedes three timed passes; the warm-up work must still be reported.

All raw passes and quality counts

Measurement	Model A	Model B
Pass 1 seconds	2.0	1.2
Pass 2 seconds	2.2	1.1
Pass 3 seconds	1.8	1.3
Correct on 100 validation cases	92	90

Repeated timings do not multiply the quality sample

The validation cases are the same for both models and are scored once each; repeating timing passes does not create 300 independent quality examples. No actual benchmark was run to produce this table.

Name given and missing facts

List the given facts and one fact that is not supplied.

Convert a whole-pass duration into two resource summaries

> Seconds per example and examples per second use opposite ratios.

Step-by-step median, time and throughput

Sort each set of three times and take the middle value. A’s median is 2.0 seconds per 100 examples; B’s is 1.2 seconds. Dividing by 100 gives amortised times of 0.020 and 0.012 seconds per example, or 20 and 12 milliseconds.

Median-pass **throughput** is 100 / 2.0 = 50 examples per second for A and about 83.33 for B.

Raw-to-derived comparison table

Derived quantity	A	B
Sorted pass seconds	1.8, 2.0, 2.2	1.1, 1.2, 1.3
Median seconds per 100 cases	2	1.2
Amortised milliseconds per case	20	12
Throughput at median pass, cases/second	50	83.33 approximately
Correct of same 100 validation cases	92	90
At least 91 correct AND at most 25 ms	Pass both	Fail quality
Timed prediction executions	300	300
Warm-up executions, excluded from timing	100	100

Original runnable timing helper

```
from math import isfinite
from statistics import median
```

```
def timing_summary(seconds, count):
    if count <= 0 or not seconds:
        raise ValueError("Need examples and timed passes")
    if any(not isfinite(t) or t <= 0 for t in seconds):
        raise ValueError("Times must be finite and positive")
    middle = median(seconds)
    ms_per_example = 1000 * middle / count
    examples_per_second = count / middle
    return middle, ms_per_example, examples_per_second

print(timing_summary([2.0, 2.2, 1.8], 100))
print(timing_summary([1.2, 1.1, 1.3], 100))
```

Different percentage denominators

The code assumes a positive integer example count and numeric times. A **output**s (2.0, 20.0, 50.0); B **output**s approximately (1.2, 12.0, 83.33). Retain full precision until display.

The time reduction is (20 - 12) / 20 = 40 percent, while **throughput** rises by about 66.67 percent. These are different denominators.

What amortised time cannot measure

Amortised time does not reveal the slowest request or a **latency** percentile. In a batched or concurrent workload, total time divided by count is not an individual request’s waiting time.

Record per-request timings if the requirement concerns **latency**; keep batch size and concurrency explicit.

Explain the pictured boundary

Point to the **input**, the deciding step and the result in this week’s visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Connect the picture and the explanation

Apply both requirements before choosing a winner

> A faster ineligible model does not pass the declared contract.

Current task and response

- With at least 91 correct out of 100 and at most 25 ms amortised time required, which model qualifies? Calculate B's time reduction relative to A and explain why the percentage **throughput** increase differs.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the changed picture

- Change: Lower the accuracy requirement to90% before a new selection exercise.

▼ **Check after predicting**

Prediction: Both become eligible; Q has the smaller measured mean amortised time.

Reason: The acceptance contract changed; Q's measured accuracy did not improve.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

Keep eligibility, selection and future evidence distinct

> Requirements are declared before inspecting the supplied results.

Main contract and evidence limits

A faster result can miss the requirement

MODEL	QUALITY REQUIREMENT	TIME REQUIREMENT
A: 92 of 100	At least 91 correct: pass	At most 25 ms average: pass
B: 90 of 100	At least 91 correct: fail	At most 25 ms average: pass

For this declared validation rule, only A meets both requirements. The time requirement uses amortised time from the median pass, not a per-request latency guarantee.

The requirements were set before inspecting these results. B is faster under this measurement but is not eligible under the quality rule.

One hundred validation cases do not establish future accuracy; reserve independent final evaluation after choosing a model, and inspect relevant subgroup results.

Try the alternative main quality requirement

> **If the quality requirement had instead been set in advance to at least 90 correct, with the same time limit, which models would qualify? Which would you choose if lower measured amortised time is the declared tie-breaker?**

Separate mean-based picture and all givens

Compare speed and quality under the same case count and acceptance contract.

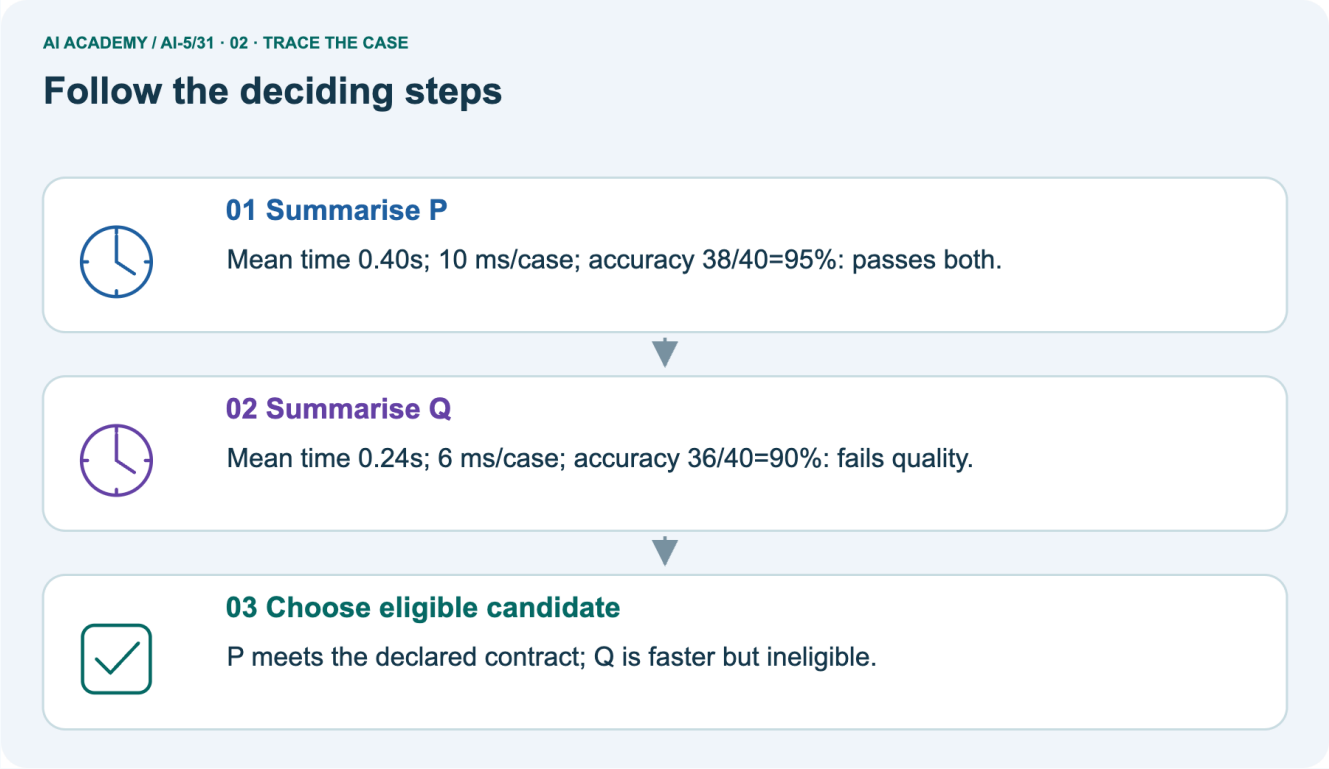
Both evaluate 40 cases. P times [0.32, 0.40, 0.48]seconds, 38 correct; Q times [0.20, 0.24, 0.28]seconds, 36 correct. Requireaccuracy>=93% and mean amortised time<=12 ms/case.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The main comparison uses the median of three 100-case passes. The picture is a separate 40-case comparison using the mean: P times .32, .40, .48 seconds average .40; Q times .20, .24, .28 average .24. That gives 10 and 6 ms per case. Accuracy P 38/40 = 95%, Q 36/40 = 90%. With at least 93% and at most 12 ms required, only P qualifies.

Although these symmetric lists also have matching medians, mean and median are different summaries in general.

Trace the picture's two requirements



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Summarise P	Mean time 0.40s; 10 ms/case; accuracy 38/40=95%: passes both.
2. Summarise Q	Mean time 0.24s; 6 ms/case; accuracy 36/40=90%: fails quality.
3. Choose eligible candidate	P meets the declared contract; Q is faster but ineligible.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

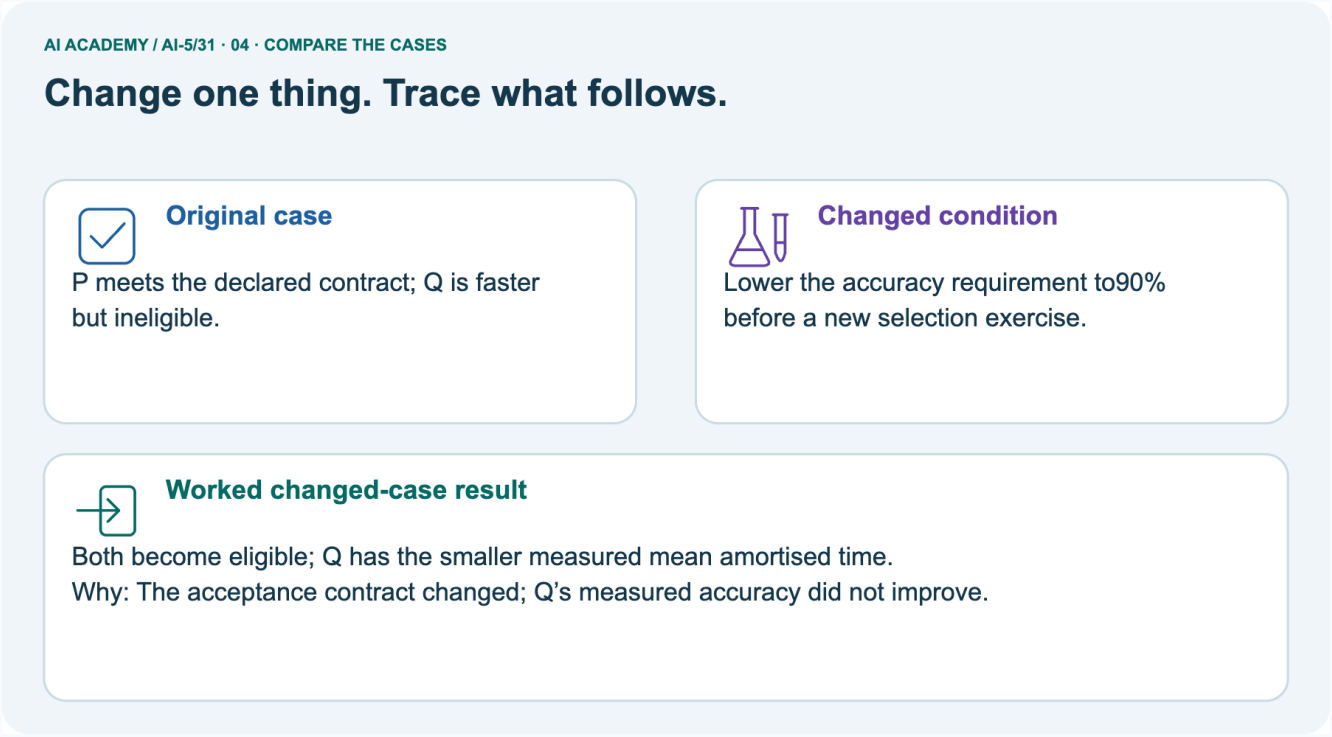
Compare models on the same workload and apply every required condition. P meets both quality and time limits; Q's speed does not compensate for missing the quality gate.

Dividing total run duration by forty gives an amortised time per case. It does not measure the wait for one isolated request, which needs its own timed experiment.

Predict the new picture requirement

- Change one thing: Lower the accuracy requirement to90% before a new selection exercise. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain amortised time and missing measurements

- Record hardware, batching, warm-up and variability; these supplied timings are toy measurements, not actual benchmark runs.

- Read the labelled picture: Why is 10 ms/case not necessarily a single-request **latency**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

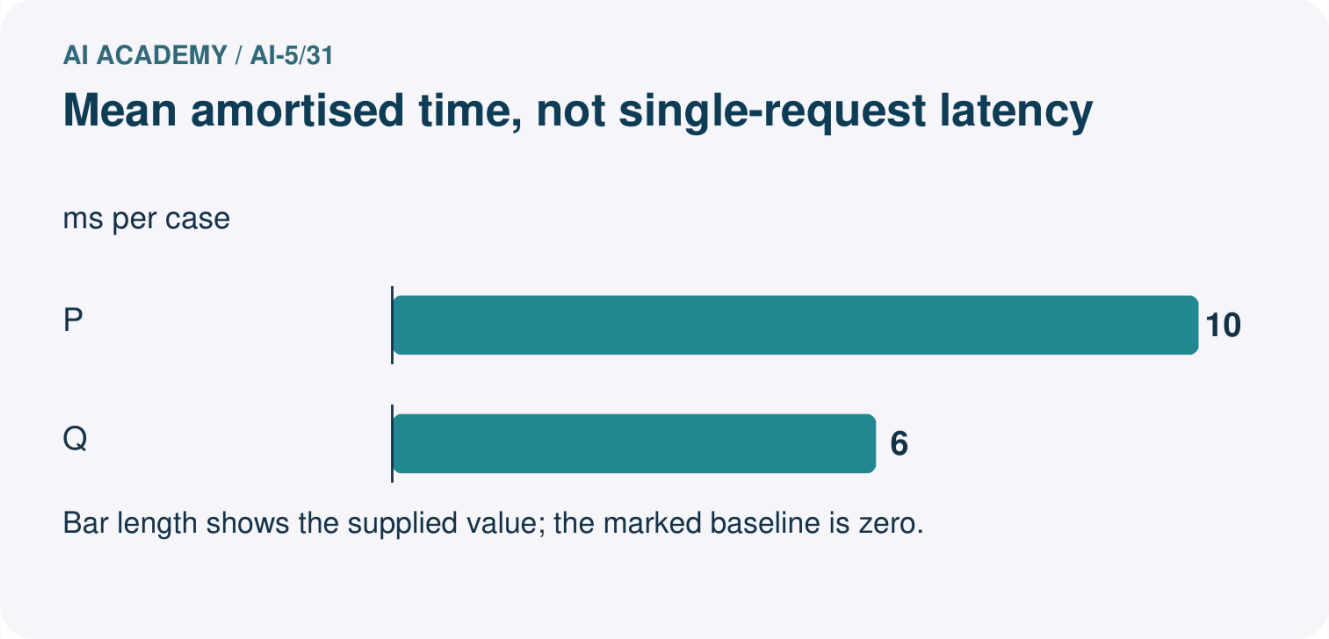
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Efficiency and resource measurement

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/31

Efficiency and resource measurement

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Summarise P

Mean time 0.40s; 10 ms/case; accuracy 38/40=95%: passes both.

↓

2. Summarise Q

Mean time 0.24s; 6 ms/case; accuracy 36/40=90%: fails quality.

↓

3. Choose eligible candidate

P meets the declared contract; Q is faster but ineligible.

CHANGE ONE THING

Lower the accuracy requirement to90% before a new selection exercise.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Count parameters without calling that peak memory

> A storage calculation and a runtime measurement answer different questions.

Dense weights, biases and raw bytes

A fully connected layer with m **input**s and n **output**s has m times n weights and n biases, assuming every **output** has one bias. A network with dimensions 4 to 3 to 2 has (4 times 3 + 3) + (3 times 2 + 2) = 23 parameters.

At four bytes per parameter, its raw parameter payload is 92 bytes.

Complete layer-by-layer account

Dense 4 → 3 → 2 network	Weights	Biases	Total parameters
4 → 3 layer	4 × 3 = 12	3	15
3 → 2 layer	3 × 2 = 6	2	8
Whole network	18	5	23
Four bytes per parameter	23 × 4 = 92 raw bytes	Metadata not counted	Activations and buffers not counted
Peak training memory	Requires actual scoped measurement	Gradients / optimizer state may add storage	Not supplied
Energy or carbon emissions	Require additional evidence	Time alone is insufficient	Not supplied

File size, runtime memory and energy limits

That is not the saved file size or peak runtime memory. A file can contain metadata, and inference also needs activations and buffers. **Training** may additionally store gradients and optimiser state.

Record whether a memory measurement refers to the model, the **process**, an accelerator or the whole device. Parameter count alone does not establish speed.

Runtime alone does not establish energy use or carbon emissions either. Those require additional measurements and declared assumptions, such as power over time and the electricity source.

Report what you measured rather than converting fewer parameters into an unsupported energy claim.

Try the resource misconception check

- Count weights, biases and total parameters in the 4 to 3 to 2 network. Calculate raw parameter bytes at four bytes each. Explain why this cannot be reported as peak **training** memory or an energy measurement.

> **State one repair and the checks you would rerun.**

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise timing ratios and quality gates

> Keep raw observations, chosen summary and decision rule visible.

Current task and alternative requirement

- For A and B, calculate median pass time, amortised milliseconds per example and median-pass **throughput**. State the measured boundary and one **latency** claim these numbers cannot support.

> **If the quality requirement had instead been set in advance to at least 90 correct, with the same time limit, which models would qualify? Which would you choose if lower measured amortised time is the declared tie-breaker?**

> **Explain your prediction to a partner before comparing results.**

Supported W2 record

Use workbook W2 for the connected example “Efficiency and resource measurement”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Parameter-count misconception

- Use the worked case for Efficiency and resource measurement. Proposed explanation: “Fewer parameters always means faster”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured latency question

- Why is 10 ms/case not necessarily a single-request **latency**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve independent timings and network

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Compare under requirements

Check your reasoning

Explain the picture

Run the complete resource and eligibility audit

> No real device benchmark is implied by calculations on supplied times.

Current eligibility and percentage investigation

With at least 91 correct out of 100 and at most 25 ms amortised time required, which model qualifies? Calculate B's time reduction relative to A and explain why the percentage **throughput** increase differs.

Complete executable audit

```
"""Arithmetic on invented timing observations; this is not a benchmark."""
from math import isfinite
from statistics import median, mean

def timing_summary(seconds, count, method='median'):
    if type(count) is not int or count <= 0 or not seconds:
        raise ValueError('Need a positive integer count and timed passes')
    if any(not isfinite(t) or t <= 0 for t in seconds):
        raise ValueError('Times must be finite and positive')
    if method not in ('median','mean'):
        raise ValueError('Unknown summary method')
    duration = (median if method=='median' else mean)(seconds)
    return duration, 1000*duration/count, count/duration

def parameters(widths):
    if len(widths)<2 or any(type(n) is not int or n<=0 for n in widths):
        raise ValueError('Need at least two positive integer layer widths')
    weights=sum(m*n for m,n in zip(widths,widths[1:]))
    biases=sum(widths[1:])
    return weights,biases,weights+biases

a=timing_summary([2.0,2.2,1.8],100)
b=timing_summary([1.2,1.1,1.3],100)
def eligible(correct,count,ms,minimum_correct,maximum_ms):
    return correct>=minimum_correct and ms<=maximum_ms
assert a==(2.0,20.0,50.0) and b[:2]==(1.2,12.0)
assert eligible(92,100,a[1],91,25) and not eligible(90,100,b[1],91,25)
assert eligible(90,100,b[1],90,25)
print('Main median summaries: seconds, ms/example, examples/second',a,b)
print('Relative time reduction:',(a[1]-b[1])/a[1])
print('Relative throughput increase:',(b[2]-a[2])/a[2])
```

```
print('Timed passes per model: 3 × 100 = 300 prediction executions')
print('Plus warm-up: 100 more executions; quality sample remains 100 distinct cases')
w,bi,total=parameters([4,3,2])
assert (w,bi,total)==(18,5,23)
print('Dense weights, biases, total, raw bytes:',w,bi,total,total*4)
p=timing_summary([.32,.40,.48],40,'mean')
q=timing_summary([.20,.24,.28],40,'mean')
assert abs(p[1]-10)<1e-12 and abs(q[1]-6)<1e-12
print('Separate picture mean summaries:',p,q)
print('Picture quality:',38/40,36/40,'required:',.93)
# No peak memory, request latency, power or carbon measurements are supplied.
```

Expected checks and measurement contract

1. Predict A/B median summaries, eligibility and dense parameter count before running or tracing. Check time reduction 40% versus throughput increase about 66.67%: both compare B with A but use different quantities and denominators. The complete helper rejects empty timings, nonpositive or nonfinite durations and noninteger or zero case counts.

   Count timed executions and warm-up separately; the distinct quality sample remains 100. Record this as arithmetic on invented observations. To benchmark a real model, specify device, precision, batch size, concurrency, completed-work boundary, warm-up and all repetitions; measure peak memory and request latency separately if required.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Reference: earlier timing function

```
from math import isfinite
from statistics import median

def timing_summary(seconds, count):
    if count <= 0 or not seconds:
        raise ValueError("Need examples and timed passes")
    if any(not isfinite(t) or t <= 0 for t in seconds):
        raise ValueError("Times must be finite and positive")
    middle = median(seconds)
    ms_per_example = 1000 * middle / count
    examples_per_second = count / middle
    return middle, ms_per_example, examples_per_second

print(timing_summary([2.0, 2.2, 1.8], 100))
print(timing_summary([1.2, 1.1, 1.3], 100))
```

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compare new timings and count a new network

> Apply the fresh quality gate before selecting an eligible model.

Complete independent timing and architecture case

For 50 shared validation cases, P has times 0.6, 0.5, 0.4 seconds and 47 correct; Q has 0.3, 0.4, 0.5 seconds and 45 correct. Require at least 92 percent accuracy and at most 12 ms amortised time.

Give median times, amortised times, **throughput**, accuracy and eligible models. A separate dense network has dimensions 3 to 2 to 1 with one bias per **output**: find its parameter count and raw bytes at four bytes each.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Evaluate a new case independently

Attempt this without hints

For 50 shared validation cases, P has times 0.6, 0.5, 0.4 seconds and 47 correct; Q has 0.3, 0.4, 0.5 seconds and 45 correct. Require at least 92 percent accuracy and at most 12 ms amortised time.

Give median times, amortised times, **throughput**, accuracy and eligible models. A separate dense network has dimensions 3 to 2 to 1 with one bias per **output**: find its parameter count and raw bytes at four bytes each.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
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> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the full comparison protocol

> Keep unsuccessful trials and warm-up in the experiment account.

Project artifact and evidence

Add a labelled efficiency and resource measurement **evidence** sheet to Image-pattern Network Investigation. Show how you can calculate **throughput** and amortised time per example from repeated timings.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	

Project evidence	Record
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run or trace and limits

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with an eligible choice and a bounded resource claim

> Do not turn amortised time into a latency, memory or energy measurement.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Calculate **throughput** and amortised time per example from repeated timings

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Latency	Elapsed time for one request from a declared start to completed response
Throughput	The number of completed examples processed per unit time
Parameter storage	The bytes used to store model weights and biases at a declared precision
Peak memory	The largest amount of memory in use during a measured operation

Use two terms to explain a step in this week's example.

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- For A and B, calculate median pass time, amortised milliseconds per example and median-pass **throughput**. State the measured boundary and one **latency** claim these numbers cannot support.

- Count weights, biases and total parameters in the 4 to 3 to 2 network. Calculate raw parameter bytes at four bytes each. Explain why this cannot be reported as peak **training** memory or an energy measurement.

- A runs on one device with batch size 1; B runs on another with batch size 32. Only B's fastest trial and the winning **training** run are reported. Explain how to compare and report the results fairly.

Explain → apply → challenge

Explain which evidence supports your conclusion and which claim remains unproved.

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the independent 50-case timings and 3→2→1 network for the first individual attempt.

Current explanation and response

- Record model and code versions, **input** size, device, precision, batch size, concurrency, warm-up, timed boundary, repetitions and raw results.

  Alternate model order where practical, check background load and wait for computation to finish before stopping a timer; queued accelerator work may finish after the host call returns.

  Report cold-start behaviour separately when it matters.

- Keep **training** and inference costs separate, and include unsuccessful trials in the total experiment budget. Changes chosen using validation need untouched final evaluation.

  Next week assembles a network experiment record so another person can understand and repeat the comparison.

- A runs on one device with batch size 1; B runs on another with batch size 32. Only B's fastest trial and the winning **training** run are reported. Explain how to compare and report the results fairly.

A report calls B “best and greener” because it is faster, compares B’s batched device run with A’s unbatched run on a different device, and omits B’s quality result. Name the unsupported claims and design a fair comparison.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

A faster result can miss the requirement

MODEL	QUALITY REQUIREMENT	TIME REQUIREMENT
A: 92 of 100	At least 91 correct: pass	At most 25 ms average: pass
B: 90 of 100	At least 91 correct: fail	At most 25 ms average: pass

For this declared validation rule, only A meets both requirements. The time requirement uses amortised time from the median pass, not a per-request latency guarantee.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Compare speed and quality under the same case count and acceptance contract.

By the end, I can	How I will show it
Calculate throughput and amortised time per example from repeated timings	Explain the deciding step, show a trace or test, and state one limit.
Count dense-network parameters and distinguish storage from peak memory	Explain the deciding step, show a trace or test, and state one limit.
Compare resource use under fixed quality and measurement conditions	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Compare speed and quality under the same case count and acceptance contract.
- Which labelled step matches this detail? P meets the declared contract; Q is faster but ineligible.
- 1. Summarise P
- 2. Summarise Q
- 3. Choose eligible candidate
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Choose eligible candidate P meets the declared contract; Q is faster but ineligible. Compare models on the same workload and apply every required condition.

P meets both quality and time limits; Q's speed does not compensate for missing the quality gate. Dividing total run duration by forty gives an amortised time per case.

It does not measure the wait for one isolated request, which needs its own timed experiment.

Changed-case question and optional hint

- Changed-case practice: Lower the accuracy requirement to90% before a new selection exercise. Which conclusion is supported by the supplied facts?
- 1. Both become eligible; Q has the smaller measured mean amortised time.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
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▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Both become eligible; Q has the smaller measured mean amortised time. The acceptance contract changed; Q's measured accuracy did not improve.

Repair and retain the record

Review Dataset shift in neural models (ai-5/30). Goal: Explain how a changed **input** mix can change overall accuracy. Short practice: Did the supplied model's dim-light accuracy deteriorate?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Transfer learning (ai-5/28). Goal: Trace a reused feature extractor and a newly fitted **output** head. Short practice: Which part was trained for the new task?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Sequence representations (ai-5/25). Goal: Encode an ordered sequence with an explicit vocabulary. Short practice: Which record lets the program distinguish the first and last zeros?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

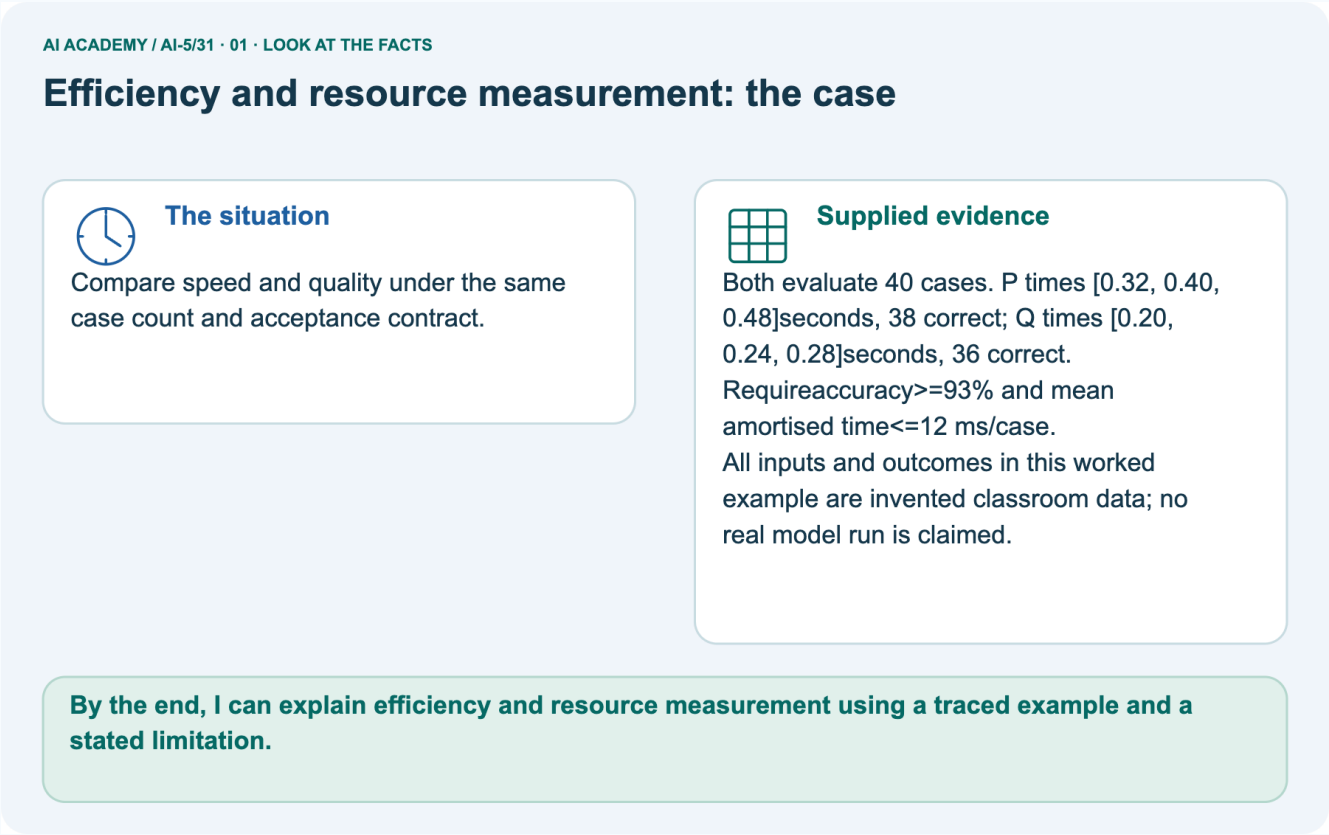
Keep your completed work

▼ Reference · Six-picture concept journey

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Efficiency and resource measurement: the case



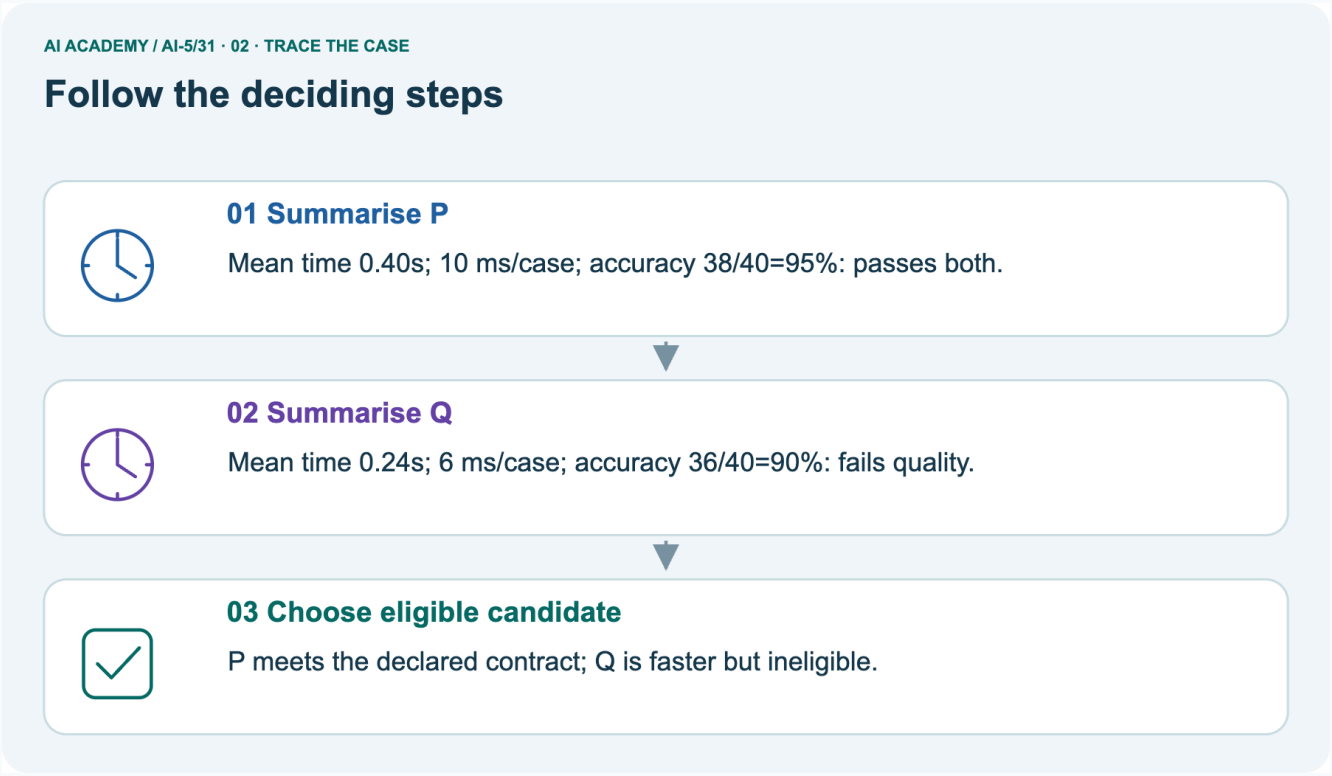
Follow the labelled picture with the matching explanation. Open the vector diagram

Compare speed and quality under the same case count and acceptance contract. Supplied facts: Both evaluate 40 cases. P times [0.32, 0.40, 0.48]seconds, 38 correct; Q times [0.20, 0.24, 0.28]seconds, 36 correct.

Requireaccuracy>=93% and mean amortised time<=12 ms/case. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

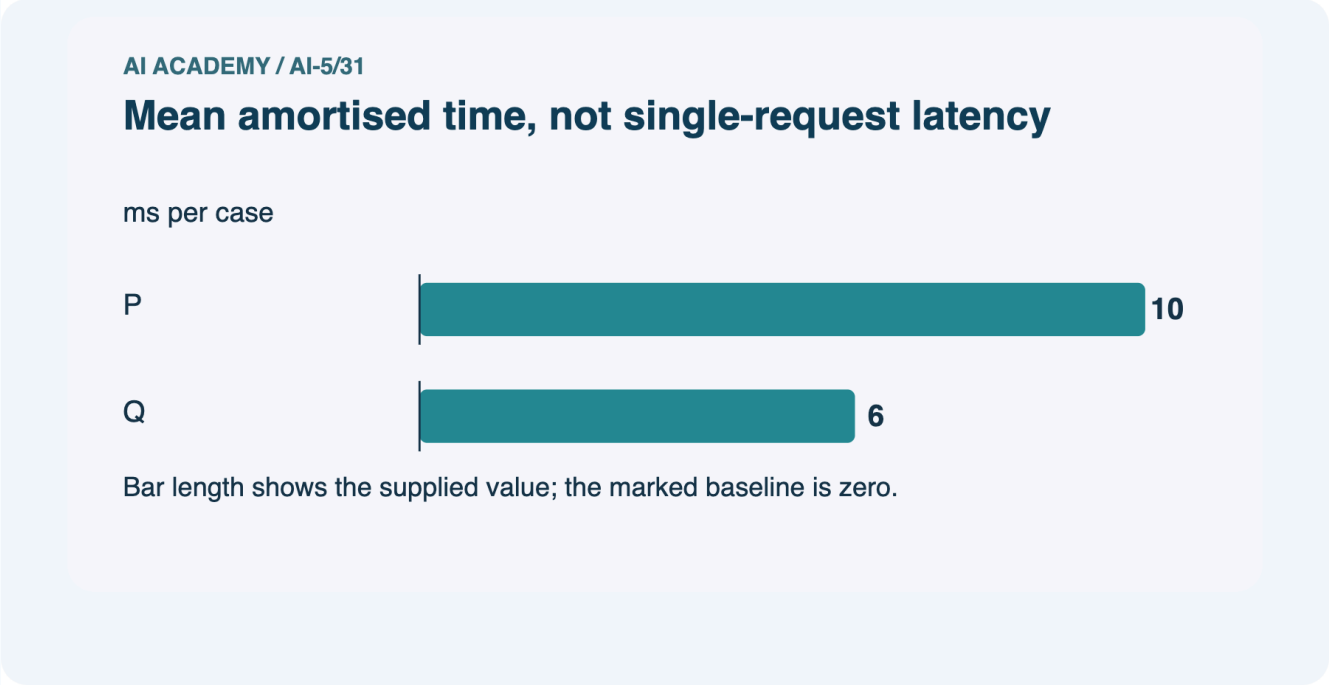


Follow the labelled picture with the matching explanation. Open the vector diagram

Summarise P: Mean time 0.40s; 10 ms/case; accuracy 38/40=95%: passes both. → Summarise Q: Mean time 0.24s; 6 ms/case; accuracy 36/40=90%: fails quality. → Choose eligible candidate: P meets the declared contract; Q is faster but ineligible.

Point to each numbered stage and explain which recorded **input** it uses.

03. Mean amortised time, not single-request latency

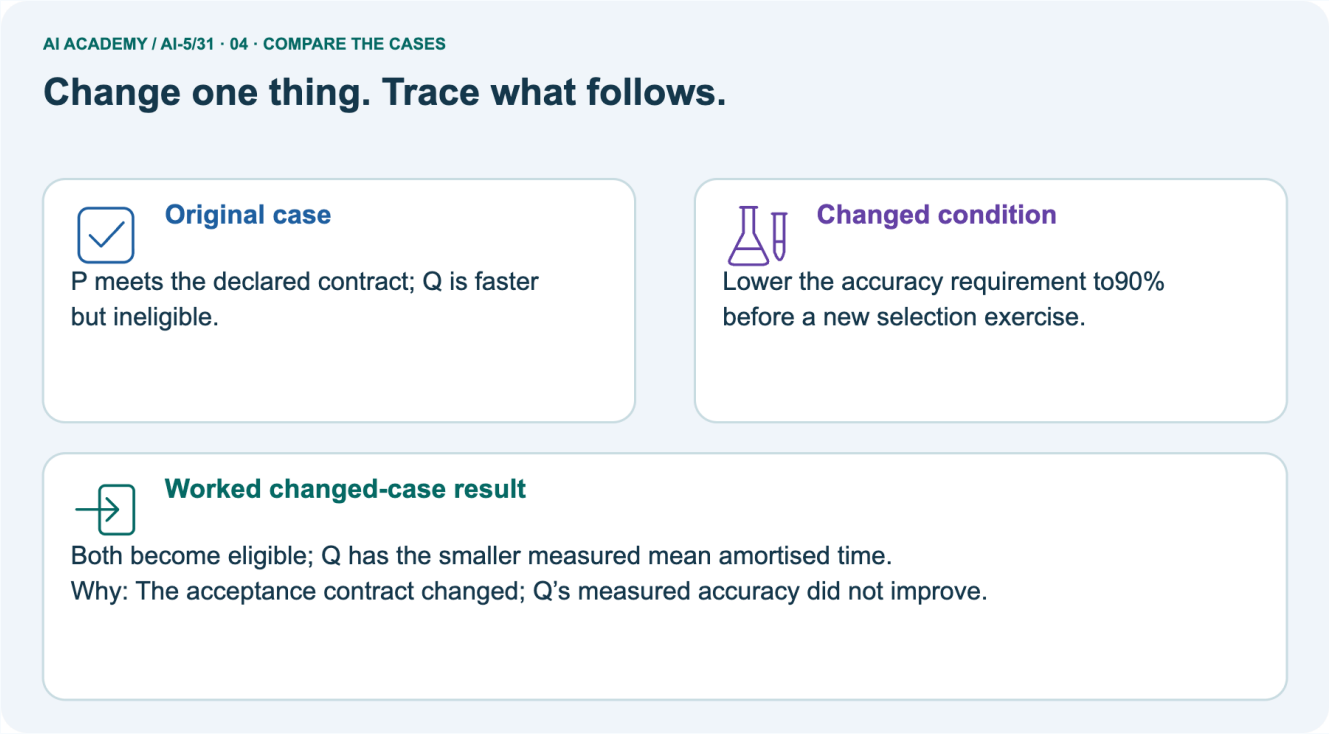


Follow the labelled picture with the matching explanation. Open the vector diagram

Mean amortised time, not single-request **latency**. Units: ms per case. P: 10. Q: 6.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



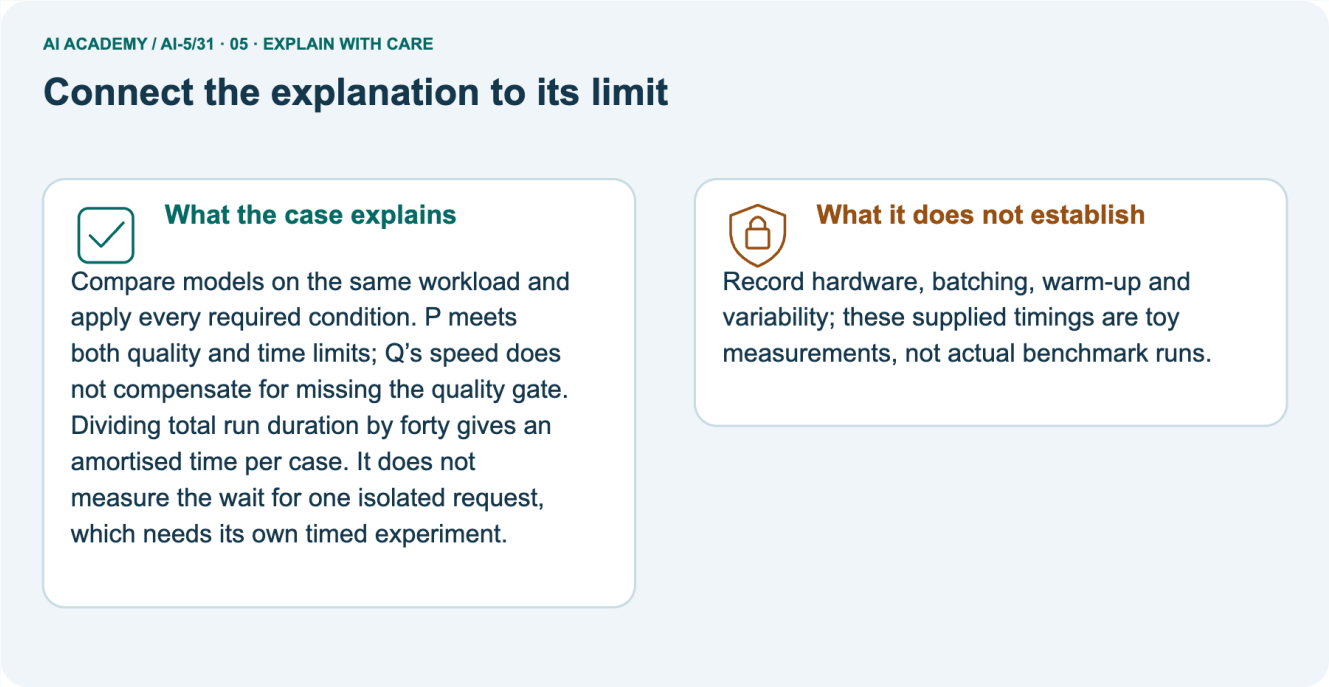
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: P meets the declared contract; Q is faster but ineligible. Changed condition: Lower the accuracy requirement to90% before a new selection exercise.

Expected result: Both become eligible; Q has the smaller measured mean amortised time. Reason: The acceptance contract changed; Q's measured accuracy did not improve.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

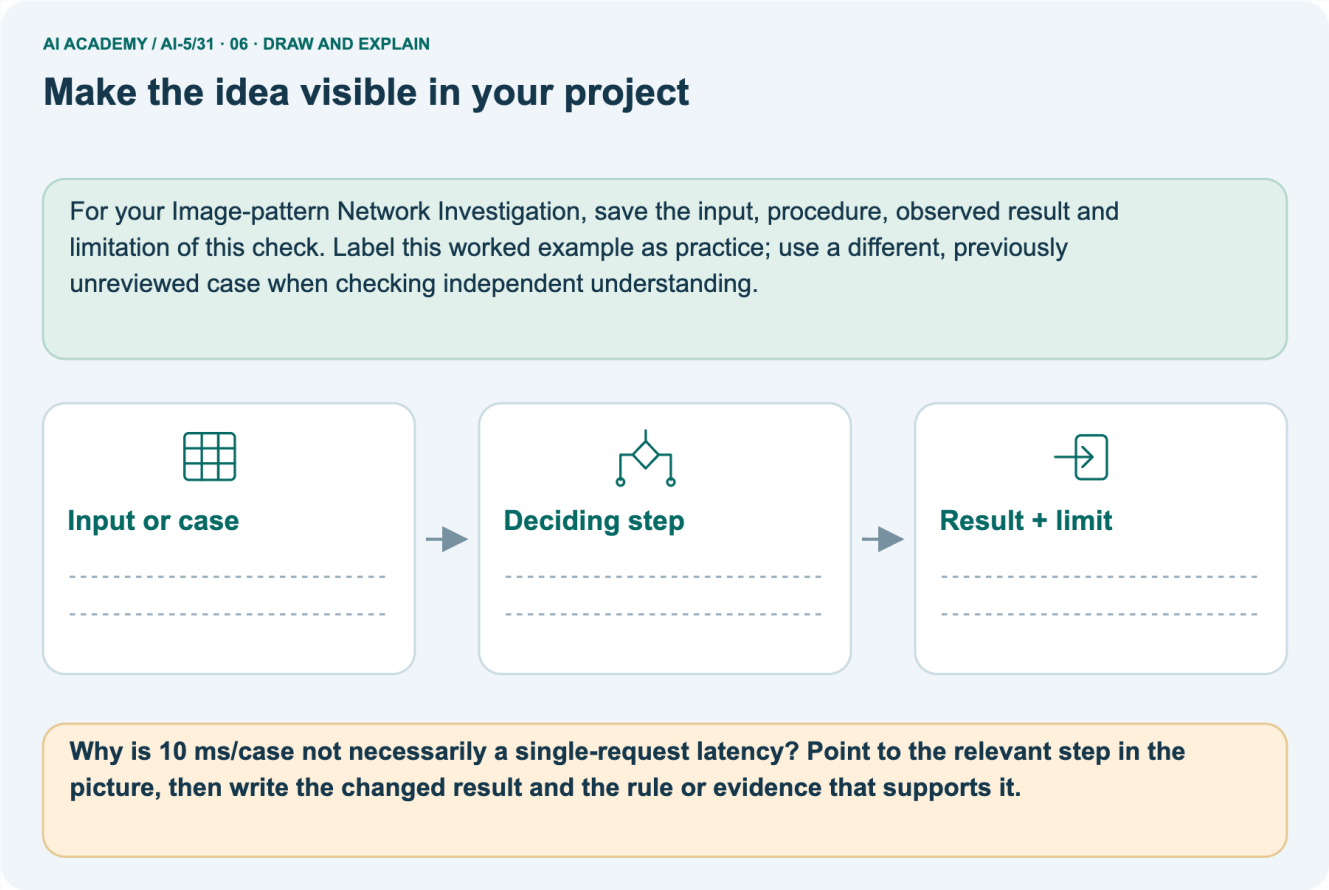
Compare models on the same workload and apply every required condition. P meets both quality and time limits; Q's speed does not compensate for missing the quality gate.

Dividing total run duration by forty gives an amortised time per case. It does not measure the wait for one isolated request, which needs its own timed experiment.

Limit: Record hardware, batching, warm-up and variability; these supplied timings are toy measurements, not actual benchmark runs.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is 10 ms/case not necessarily a single-request **latency**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Efficiency and resource measurement: the case

02. Follow the deciding steps

03. Mean amortised time, not single-request latency

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.